

Natalie Leaist

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EDUCATION

The University of British Columbia, B.Asc. in Engineering Physics

Vancouver, BC | Sept. 2023 - Present

- 94.6% cumulative average (4.33/4.33 GPA)
- Trek Excellence Scholarship recipient (top 5% of faculty and year), EnSURE Award recipient & Dean's Scholar
- Pursuing minor in Honours Math

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, Javascript

Embedded Systems: STM32, TMS570, FreeRTOS, I2C, SPI, ethernet (UDP, lwIP), ADCs, sensors, Arduino

Data Analysis & Modelling: NumPy, matplotlib, Jupyter, FFTs, basic statistical tests, MATLAB, Simulink

Electrical: KiCad, PCB design, soldering, oscilloscope, function generator, multimeter

Machining: 35 Hours of training in machining strategies and techniques, including mill, lathe, waterjet cutter, drill press and hand tools

Other Tools: CAD (SolidWorks, OnShape), Git, Github, GitLab, VSCode, IntelliJ, Linux/WSL, OpenCV, LaTeX, Microsoft Office

EXPERIENCE

Research Intern, UBC Physics & Astronomy

Vancouver, BC | May 2025 – Present

- Worked on the CGEM project, a radio telescope that will be used to help detect gravitational radiation from the early universe (collaboration between UBC, NASA Goddard Space Flight Centre and DRAO)
- Developed a high-frequency (830 Hz) embedded data acquisition system for accelerometer-based vibration and inclination monitoring of the telescope using an STM32 microcontroller, AD7177-2 ADC, DMA, SPI communication, FreeRTOS, and Ethernet (UDP/LwIP) streaming
- Performed Python-based data analysis, including FFTs, to verify accelerometer performance and calibrate sensor offset/gain
- Created a calibration procedure for accelerometer offset and gain, enabling accelerometers to have displacement resolution of <1 micron
- Prepared a detailed 25-page technical report in LaTeX

Firmware & Control Systems Developer, UBC Orbit Satellite Design Team

Vancouver, BC | Sept. 2024 - Present

- Working on the ALEASAT project, an Earth observation satellite supported by the European Space Agency's Fly Your Satellite! Program
- Developed firmware and software to fulfill mission and testing requirements set by the European Space Agency
- Implemented CPU usage monitoring across 15 FreeRTOS tasks, increasing system reliability
- Working on a team of five to develop a Helmholtz cage to simulate the magnetic field of low Earth orbit (LEO)
- Developing a MATLAB/Simulink model of the Helmholtz cage control system to simulate and ensure accurate magnetic field generation

Outdoor Adventure Guide, The Adventure Group

Whistler, BC | July 2022 – Aug. 2023

- Led groups of up to 20 guests on Canada's highest and 2nd longest zip-line
- Received over 50 five-star guest reviews, the highest among guides, highlighting strong communication and leadership
- Conducted safety inspections on off-road vehicles, zip-lines, and aerial obstacle courses
- Trained in leadership, emergency response, risk assessment, and guest management

PROJECTS

Self-Balancing Robot

- Designed and built a two-wheeled self-balancing robot using Arduino, MPU6050 IMU, DRV8825 motor drivers, and NEMA-17 stepper, gaining hands-on experience in hardware integration and embedded programming
- Implemented a complementary filter to combine accelerometer and gyroscope data, providing a stable pitch estimate and developing practical skills in sensor fusion and noise reduction
- Developed and tuned a PID controller to actively stabilize the robot's pitch

Wordle Solver

- Developed a Python algorithm capable of solving Wordle puzzles with an average of approximately 4 guesses
- Implemented a word-elimination strategy to reduce the number of possible solutions after each guess
- Used object-oriented programming in Python to organize the solver into modular, reusable classes

Rubik's Cube Solver (in progress)

- Developed an OpenCV-based computer vision script in python to detect and map Rubik's Cube face stickers from images with high accuracy, gaining practical experience in image processing
- Implementing solving algorithms using group theory concepts (commutators, conjugation) to generate efficient move sequences

ACTIVITIES

Freeride Skiing

- Ranked 2nd in Canada in my age category (2022-2023 season), qualifying for Junior World Championships (2023) as one of ~15 junior female riders worldwide
- Recognized with Rossignol Freeride Platinum Sponsorship (Sept. 2023 – Aug. 2024), the highest athlete tier before Rossignol's professional team

CERTIFICATIONS

- Class 5 Drivers' License
- 16 Hour Standard First Aid
- PMBIA Level 1 Mountain Bike Instructor
- Level 2 Avalanche Skills Training